INCH-POUND

MIL-B-43998D
17 November 1992
SUPERSEDING
MIL-B-0043998C(GL)
13 August 1990
and
MIL-B-43998B
27 July 1983

MILITARY SPECIFICATION

BATH UNIT, AUTOMATED, MULTI-HEAD

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable field bath unit capable of supplying heated water to nine shower heads. The water heater supplied with the unit will heat water to 100°F rise at a rate of 9 gallons per minute and operates on JP-8 fuel or diesel fuel. An alternating current 208 volts electric power is required.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5017 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4510

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SPECIFICATIONS

FEDERAL

TT-W-572	-	Wood-Preservative: Water-Repellant
VV-F-800	-	Fuel Oil, Diesel
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-60	-	Tape: Packaging, Waterproof

MILITARY

MIL-P-116	-	Preservation, Methods Of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-T-704	-	Treatment and Painting of Materiel
MIL-H-44086	-	Heater, Water, Liquid Fuel; M-80
MIL-C-46168	-	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-T-83133	-	Turbine Fuels, Aviation, Kerosene Types, NATO F-34 (JP-8)

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those listed in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

6-1-8198	-	Finish Requirements for Bath Unit, Automated, Multi-Head and Heater, Water, Liquid Fuel, M-80
6-1-8199	-	Bath Unit, Automated, Multi-Head
6-1-8211	-	Pump Performance Curve

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6-1-8300 - Container No. 1. Bath Unit, AMH
6-1-8301 - Plate, Packing Instructions, Container No-1; Bath Unit, AMH
6-1-8320 - Container No. 2. Bath Unit, AMH
6-1-8321 - Plate, Packing Instructions, Container No-2; Bath Unit, AMH

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN WELDING SOCIETY (AWS)

Standard Qualification Procedure of the American Welding Society

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Welding Qualifications of the American Society of Mechanical Engineers

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electric Code

(Application for copies should be addressed to National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269.)

HYDRAULIC INSTITUTE STANDARDS

(Application for copies should be addressed to the Standard Hydraulic Institute, 122 East 42nd Street, New York, NY 10017.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the bath unit to meet the performance requirements of this specification. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3 Alternate components. Components offered as equivalent to components specified herein and on the referenced drawings as a specific manufacturer's part number or equal shall be functionally equal, of equal or better quality with the manufacturer's part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the first article sample, or if none is required, prior to commencing production, the contractor shall submit for the contracting officer's approval, a list identifying each alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.4 Design and construction. The design and construction of the bath unit shall be as specified herein, and as shown on Drawing 6-1-8199 and all subsidiary drawings and parts list pertaining thereto. The water pump shall operate on a 3-phase, 60 hertz, 208 volts, 5 wire electric power source. Three wires are to supply the phases, the white wire is to be 115 volt neutral ground and the green wire is to be the equipment ground. All wiring shall be in accordance with the provisions of The National Fire Protection Association (NFPA) No. 70. Diesel shall be the primary fuel for operation of the heater and shall be in accordance with VV-F-800. JP-8 fuels shall be the

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secondary fuel and shall be in accordance with MIL-T-83133. Surfaces shall be prepared, primed and enameled after fabrication as specified on Drawing 6-1-8198. Reusable containers No. 1 and No. 2 shall have attached a photo-sensitive aluminum packing instruction label with black characters on natural aluminum background. The label for container No. 1 shall be a reduced 10 x 12 inch copy of the information on Drawing 6-1-8301. The label for container No. 2 shall be a reduced 10 x 12 inch copy of the information on Drawing 6-1-8321.

3.4.1 Heater, water. The water heater of the bath unit shall be in accordance with Type I, M-80 Water Heater, of MIL-H-44086. When specified (see 6.2) a sample water heater shall be subjected to the first article inspection requirements of MIL-H-44086.

3.5 Performance.

3.5.1 Valve, mixing. The mixing valve shall be adjusted so that the mixed water temperature is $105^{\circ}F \pm 5^{\circ}F$ and the valve shall automatically shut off the flow or outlet water when the cold water flow to the valve is terminated. Testing shall be as specified in 4.4.5.

3.5.2 Unit assembly. The complete bath unit shall be assembled and put in operation without the use of tools, except for a 7/8 inch wrench to attach the fuel lines to the drum fill assembly. Testing shall be as specified in 4.4.5.

3.5.3 Water flow. The assembled bath unit shall be capable of supplying and maintaining water to any one or all of the shower heads when tested as specified in 4.4.5.

3.5.4 Leakage. The assembled bath unit shall show no evidence of leakage when tested as specified in 4.4.5.

3.5.5 Pump, water. The water pump performance curve shall equal or exceed that shown on Drawing 6-1-8211. The net positive suction head shall not be more than 4 feet, and the flow rate, when operating at the rated motor amperage, shall not be less than 30 gallons per minute (GPM) (see 4.4.1.1). Testing shall be as specified in 4.4.5.

3.6 Marking. Marking for identification shall be in accordance with MIL-STD-130.

3.7 Interchangeability. All component parts of any one bath unit shall be so fabricated as to be interchangeable with the like parts of any other unit furnished under any one contract.

3.8 Workmanship. The bath unit components shall be free from fractures, splits, punctures, tears, dents, creases, bows, miscasts, deteriorations or malformations. There shall be no sharp edges, burrs or slivers. The finish

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shall be free of blisters, peel, chipped areas or areas of no film. The finish shall be free from smudges, poor touch up, scale, scratches, stain, overspray, sag, imprint or embedded foreign matter. All bolts, studs, nuts, and screws shall be drawn tight.

3.8.1 Welders. Unless otherwise specified (see 6.2), before assigning any welder to work, the contractor shall provide the Government inspection service with the names of the welders to be employed in the work, together with certification that each of the welders has passed a qualification test as prescribed by either of the following listed codes for the type of welding operation to be performed, and that such qualification is effective as defined by that particular code as follows:

(a) Standard Qualification Procedure of the American Welding Society.

(b) Welding Qualification of the American Society of Mechanical Engineers.

3.8.2 Welding. The surface of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be sound, continuous, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits (when flux is used) shall be removed from the finished welds.

3.8.3 Electrical wiring. Wiring shall not be cut, abraded, or have excessive insulation stripped, and shall be properly and tightly joined at terminals. Wiring shall have adequate slack to provide strain relief.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling

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inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and shall be tested for the characteristics specified in 4.4.5. Any nonconformance shall be cause for the rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the water pump conforms to the requirements specified in 3.5.5 and certification of welders as specified in 3.8.1 and NFPA wiring be acceptable on a certificate of compliance.

4.4.2 In-process inspection. Examination shall be made of surface preparation prior to application of the finished coat to determine compliance with the requirements of Drawing 6-1-8198. Examination of the surface preparation prior to welding shall be made to determine compliance with 3.8.2. Whenever nonconformance is noted, correction shall be made to the applicable process and all items processed. Items which cannot be corrected shall be removed from production.

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4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of completely assembled bath units. The sample unit shall be one completed bath unit. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Classification Minor
Finish	Color not as specified		201
	Blistered, peeled, chipped, or area of no film		202
	Poor touch-up, orange peel, color separation, smudge, scale, scratched, stained, overspray, sag, imprint, or foreign matter imbedded		203
Design, construction and workmanship, general (applicable to all components and assemblies)	Any component or part missing or not specified type or size	101	
	Fractured, split, punctured, torn, dented, creased, deteriorated, or malformed	102	
	Component misplaced or not in proper alignment		204
	Component loose		205
	Sharp burr or edge or sliver		206
	Any operation omitted or not properly performed	103	
	Any functioning component that is inoperative or will not operate as intended	104	
	Any functioning component that requires abnormal force to operate or is not interchangeable		207
	Any adjustable assembly that cannot be adjusted		208
	Any assembly not properly adjusted		209
	Any component not readily accessible for servicing where required		210
Welding	Missing, incomplete, cracked, fractured, not fused	105	
	Undercut, not smooth and uniform, or scale or flux deposits not removed		211

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Soldering (when applicable)	Missing, not adherent, or incomplete		212
	Not clean (flux residue not removed, pinholes in solder)		213
Bolts, nuts, screws, studs and other types of threaded fasteners	Broken, stripped, fractured, or not tight		214
Rivets (when required)	Broken, not peened or insufficiently peened, or loose		215
Electrical assembly	Wiring cut, abraded, or not properly installed or joined	106	
	Adequate slack not provided for wiring to relieve strain		216
Gauges, instruments, controls, thermometer, etc.	Not properly mounted or not specified type or capacity		217
	Not free of foreign matter		218
Marking	Missing, incomplete, not legible, not specified type or size, or not in proper location		219

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of completely assembled bath units. The sample unit shall be one completed bath unit. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. Each end item shall be tested for the characteristics listed in table II. Failure of any test shall be cause for rejection of the end item involved.

TABLE II. End item tests

Characteristic	Requirement	Test
Mixing valve	3.5.1	4.5.1

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TABLE II. End item tests (cont'd)

Characteristic	Requirement	Test
Unit assembly	3.5.2	4.5.2
Water flow	3.5.3	4.5.3
Leakage	3.5.4	4.5.4
Water pump	3.5.5	4.5.5

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE III. Packaging defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Preservation	Improperly applied or missing
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as: incomplete closure of barrier materials, not taped as specified System not completely drained and dried

4.5 Methods of inspection.

4.5.1 Mixing valve test. At water flows of 5, 8 and 12 gallons/minute the mixed water temperature shall be maintained at $105^{\circ}F \pm 5^{\circ}F$. When the hot water flow to the mixing valve is closed the outlet flow shall be reduced to less than 1 gallon/minute within 15 seconds. Any nonconformance with the mixing water temperature and valve shut off requirements of 3.5.1 shall constitute failure of this test.

4.5.2 Unit assembly test. One complete unit shall be assembled and placed in operation. Any nonconformance with the assembly and operation requirements of 3.5.2 shall constitute failure of this test.

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4.5.3 Water flow test. With the complete system in operation, each shower head shall be opened individually in one minute intervals until all are functioning. Then at one minute intervals each shower head shall be closed. Nonconformance with the water flow distribution to the shower head as specified in 3.5.3 shall constitute failure of this test.

4.5.4 Leakage test. One end of the hose assembly shall be capped, the other end shall have 50 psi water pressure applied for five minutes. The hose assemblies shall be checked for leaks. Any nonconformance with the leakage requirements of 3.5.4 shall constitute failure of this test.

4.5.5 Water pump. The water pump and motor shall be tested in accordance with the applicable pump test code of the Hydraulic Institute (see 3.5.5).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation.

5.1.1.1 Drainage of water system. The entire water system of the bath unit, including the water heater, water pump, water suction strainer, mixing valve, shower stands, water hoses, etc., shall be thoroughly drained, blown out and dried. All valves and draincocks shall be left open.

5.1.1.2 Fuel system. The fuel supply and return lines shall be connected to the oil filter and fuel pump, respectively. The hose shall be coiled beneath the heater and the ends connected to the fuel line holder. The burner pump shall be filled with grade DF-1, diesel oil of VV-F-800.

5.1.1.3 Cleaning, drying, and preservative application. All uncoated and unpainted surfaces of the bath unit shall be cleaned in accordance with MIL-P-116, process C-1, and thoroughly dried prior to application of preservative. Preservative specified in 5.1.1.5 and 5.1.1.6 shall conform to the applicable specification listed in MIL-P-116. Barrier material specified in 5.1.1.5, 5.1.1.6 and 5.1.1.11 shall conform to type I or II, grade A, class 2 of MIL-B-121. Tape specified hereinafter shall be black, conforming to type IV, class 1 of PPP-T-60.

5.1.1.4 Repainting and recoating. Painted and permanently coated surfaces from which the paint or permanent coating has been removed or which are defective, shall be repainted or recoated with material identical with the original.

5.1.1.5 Water pump assembly. The drained and dried water pump shall be preserved by spraying or fogging the interior with P-10, type I, grade 30 preservative through the suction and discharge ports while the pump is slowly actuated. Spraying or fogging shall continue long enough to insure complete

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coverage. All water line openings on the pump shall be sealed with tape. All openings on the motor shall be sealed with barrier material and tape, or by tape only, as applicable.

5.1.1.6 Water heater assembly. The water heater assembly shall be drained and dried of water and fuel as specified in 5.1.1.1. All interior surfaces that are unlined, or not treated to resist corrosion, shall be coated with P-10, type I, grade 30 preservative applied by fogging, spraying, or completely filling and draining. Excess preservative shall be drained and the drain and fill plugs, before being reinstalled, shall be coated with the preservative. All water line openings on the water heater assembly shall be sealed with tape. All openings on the air shutter, smoke stack, and electric motor shall be sealed with barrier material and tape, or by tape only, as applicable. All gauges, dials, and sight glasses not protected by a cover, shall be cushioned, overwrapped with barrier material, and the wrap secured in place with tape.

5.1.1.7 Shower stand assembly. The drained and dried shower stand assembly shall be disassembled by removing the legs and connecting water hoses. All open ends of the legs, header pipes, and brackets shall be sealed with tape. All shower heads shall be cushioned, overwrapped with barrier material, and the wrap secured in place with tape.

5.1.1.8 Water hose assemblies. The openings of the 1 inch by 7.5 feet and the 1.5 inch by 6 feet water hoses shall be sealed with tape. The 1 inch by 25 feet hoses shall be individually coiled with the ends clamped together.

5.1.1.9 Power cable assembly. The receptacle ends of the electric cords shall be sealed with tape. The cable shall be coiled and secured with tape.

5.1.1.10 Windbreakers. Each windbreaker shall be neatly and evenly folded.

5.1.1.11 Mixing valve assembly. The drained and dried mixing valve shall have all water line openings sealed with tape. The temperature gauge shall be cushioned, overwrapped with barrier material, and the wrap secured with tape.

5.1.1.12 Drum fill adapter assembly. The drum fill adapter extension shall be threaded, hand-tight into the 1/4 inch return port.

5.1.2 Commercial preservation. The bath unit shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or Commercial, as specified (see 6.2).

5.2.1 Level A packing.

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5.2.1.1 Water heater assembly and lower smoke stack assembly. The water heater, preserved as specified in 5.1, shall be packed in a snug-fitting nailed wood box constructed in accordance with class 2, grade A, style 7, type 3 load of PPP-B-621. The water heater shall be bolted to the box bottom through the skids. Additional skids are required and shall be located on the box bottom adjacent to the inner edges of the outside skids to allow for bolting of the water heater through the skids. The elbow and lower stove pipe section of the smoke stack assembly shall be attached to the heater. Assembly of the box shall be in accordance with the appendix of PPP-B-621, except that the sheating on the ends shall be attached to the interior frame members with nails and to the side of the skids with screws. The exterior and interior of wood surfaces shall be cleaned and treated in accordance with the surface application paragraph of MIL-T-704, using composition B of TT-W-572. The exposed metal surfaces shall be cleaned and treated in accordance with type F or type G of MIL-T-704. All exterior surfaces shall be painted Camouflage Green 383, conforming to type II of MIL-C-46168. The word "CARC" and the month and year of application shall be stencilled in 1 inch block letters, using lusterless black paint conforming to type II of MIL-C-46168, on the outside of the container directly below the marking required in 5.3.1. The assembled and painted box shall be strapped as specified in the appendix of PPP-B-621.

5.2.1.2 Bath unit components other than water heater assembly. The bath unit components, other than the water heater, preserved as specified in 5.1, shall be packed in the appropriate container in accordance with the instructions shown on Drawings 6-1-8300 and 6-1-8320. Cushioning material shall be applied as required to prevent movement of the components within the containers. Each packed and closed container shall be fitted with a snug-fitting fiberboard cap, 6 inches (± 1/8 inch) in depth. The cap shall be fabricated from fiberboard conforming to type CF or SF, grade optional, class weather-resistant of PPP-F-320. Caps shall be secured by applying three non-metallic straps. Two straps shall be applied girthwise, over the sides, top and bottom, and positioned to divide the distance between the ends of the container into units of equal length. The third strap shall be applied lengthwise, and centered over the ends, top, and bottom.

5.2.2 Commercial packing. The bath unit preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951. Container number 1 (Drawing 6-1-8300) and container number 2 (Drawing 6-1-8320) are components of the bath unit and are to be supplied and packed with the required equipment the same as specified for level A packing.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.3.1 Additional marking. For level A shipping containers as specified in 5.2.1.1, the following additional marking shall be stencilled on the top of the container in 1 inch high block letters using lusterless white paint conforming to type II of MIL-C-46168:

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REUSABLE CONTAINER

DO NOT DESTROY

TO OPEN REMOVE SCREWS

5.4 Depreservation guide. A depreservation guide shall be prepared, placed in a waterproof envelope marked "Depreservation Guide", and secured in a conspicuous location on the bath unit. Unless otherwise specified (see 6.2 and 6.4), DA Form 2258-Depreservation Guide for Vehicles and Equipment shall be used.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The portable bath unit is a self-contained unit and is intended for use in the field by military personnel. The unit is designed to take water from streams, rivers, lakes and the like by means of a pump,then disinfect and heat it sufficiently to provide a normal bath water temperature from the shower heads.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. When a first article is required for the water heater (see 3.4.1).
e. When a certification of qualification test for welders is not required (see 3.8.1).
f. Levels of preservation and packing (see 5.1 and 5.2).
g. When other than DA Form 2258 is to be used (see 5.4).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 DA Form 2258. The contracting officer should arrange to furnish DA Form 2258 when requested by the contractor.

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6.5 Subject term (key word) listing.

Clean
Cleaning
Portable
Shower heads
Wash
Washing
Water heater

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD, ME, TR
Air Force - 84
DLA - CS

User activity:

Navy - MC

Preparing activity:

Army - GL

(Project 4510-0278)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-43998D	1992 November 17

3. DOCUMENT TITLE

BATH UNIT, AUTOMATED, MULTI-HEAD

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4531	256-4531
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-UXT Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete. 198

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